name: ______

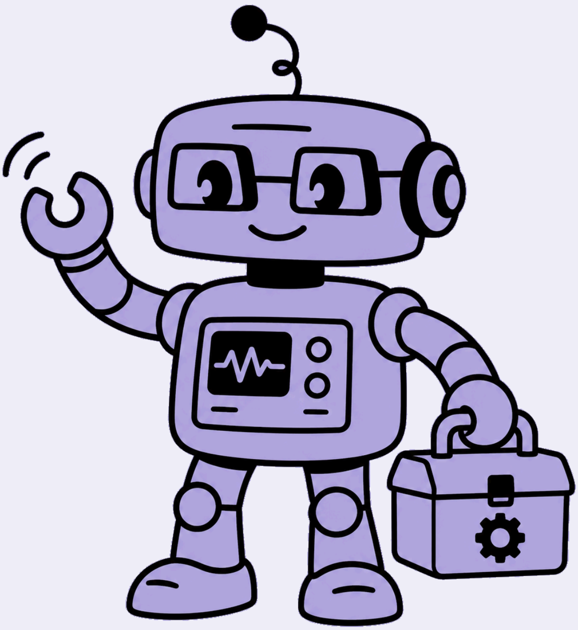
openhouse

robotics · level 2

electronics

your experience & progress book

ages 8-12



the skills you grow

in every machine you grow the same four skills. look for the little icon at the top of each page – it tells you which skill you're using.

🔧 **building & making**
you build a real machine from blocks and circuit cards, step by step.

🔬 **observing & understanding**
you watch what your circuit does – then work out *why* it works.

🧩 **problem solving**
you think up and design your own ways to make it better.

🗣️ **presenting**
you present your machine to the group, walk through how it works, and answer their questions.

what's inside

table of contents

each machine takes you through all four skills – build it, watch it, understand it, improve it, and present it.

before you begin

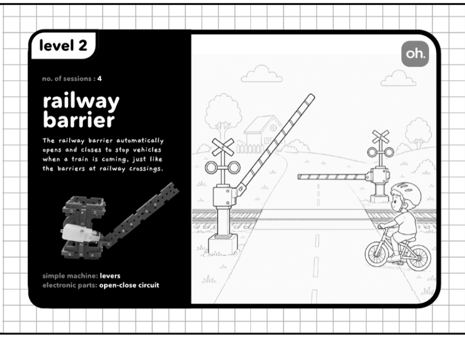
staying safe with electricity

electricity is a helper, but we treat it with care. four rules for every engineer:

- the battery blocks in your kit are **safe** to touch and build with – never use wall plugs or house electricity.
- never put wires or blocks in your mouth, or near water.
- if a wire feels warm, switch it off and tell your teacher.
- always build with your teacher nearby – engineers work safely.

for the child

meet the railway barrier



your railway barrier – it lifts and lowers when you flip the switch.

Every morning, Aarav's school bus stops at the railway crossing. A striped barrier comes down, a bell rings, and everyone waits as the train rushes past. Then the barrier lifts and the bus rolls on. how does it know when to go up and down? today, think like an engineer.

think about it

have you ever waited at a railway crossing? what made the barrier go up, then come down? draw it, or tell a friend.

words to know

circuit · switch · electricity

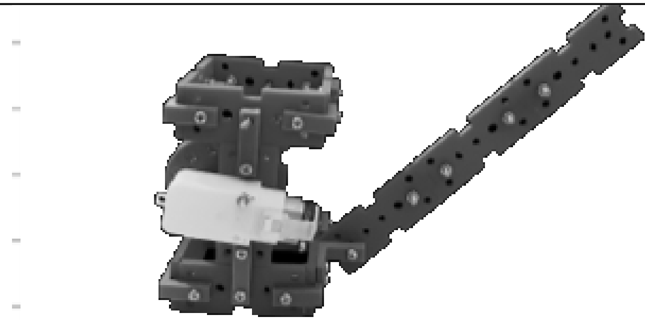
did you know?

the **electricity** flows round the loop to make it work – but only when the loop is joined all the way round.

for the child

🔧 building & making

build your railway barrier.



this is what you're building.

📖 **use your model manual**

follow your **railway barrier manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: add the barrier arm and the switch, so it **lifts and lowers when you flip the switch**.

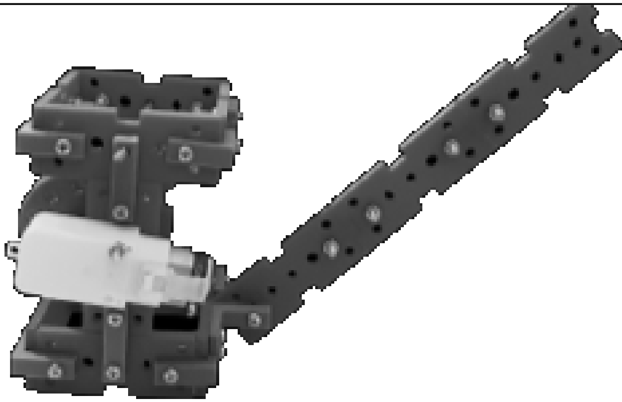
railway barrier

for the child

🔧 building & making

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

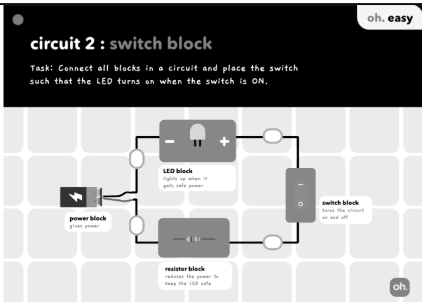
the motor | the battery | the switch | the barrier arm

⭐ **bonus:** draw an arrow to show how it moves.

for the child

observing & understanding

build this circuit – then test it like a scientist.



your test

test it 5 times. each time you close the switch, does the light come on? record it.

try	switch closed?	light on?
1		
2		
3		
4		
5		

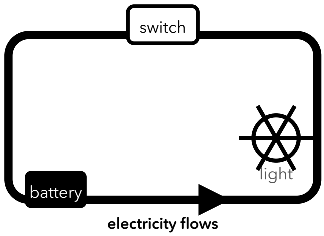
✎ what does your table show? write one sentence.

✎

for the child

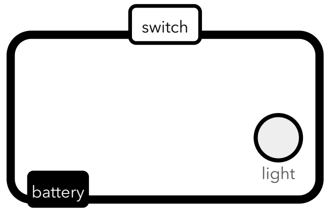
observing & understanding

the big idea – why does it work?



the loop is joined

☐



the loop has a gap

☐

a switch is simply a **gap in the loop that you control**. loop joined = a **closed circuit** → current flows. a gap = an **open circuit** → no current. current only ever flows in a complete, closed loop.

✎ in your own words

explain the big idea to a friend – write it in one or two sentences.

✎

for the child

problem solving

design challenge

real engineers try more than one idea, then compare.

1 · design a barrier where ONE switch controls TWO lights
draw a circuit that does this.

2 · now design a second way to do the same thing
draw it below.

3 · compare your two designs
which one uses less wire? circle it and say why.

✎ draw and label your designs here

for the child

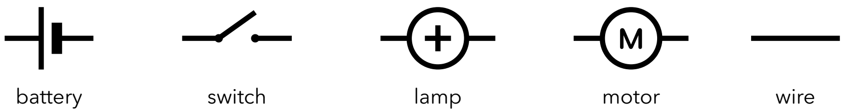
railway barrier

problem solving

⭐ engineer challenge – draw the circuit diagram

draw a **circuit diagram** (using the symbols above) for a barrier light controlled by a switch.

use these circuit symbols

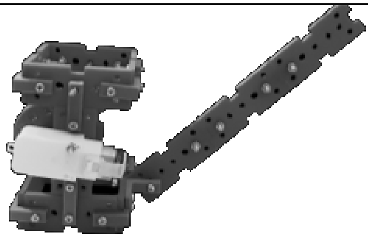


✎ draw your circuit as a proper diagram – a **complete loop** using the symbols above.

✎ draw your circuit diagram here

presenting

be the engineer – present your machine



show your railway barrier to the group and take them through it:

1 **present** – present your railway barrier to the group – show it working and name what each block does.

2 **walk through** – walk them through how it works – trace how the electricity flows, step by step, so they can follow.

3 **justify** – explain **why** it's built this way – what each block adds, and what would happen if you removed or swapped one.

4 **field it** ⭐ – take a question from the audience – answer a 'what if…?' – or coach a friend to build or fix the same circuit.

at-home show & tell

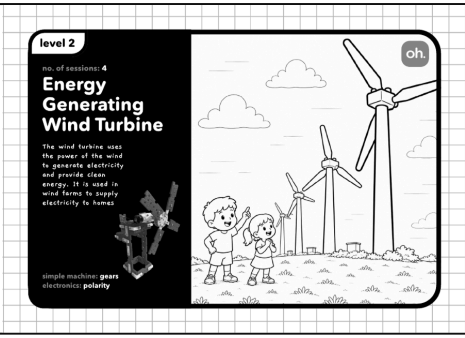
take it home and show a family member. explain: **what problem does it solve? how does it work?** (it keeps cars and people safe at the railway crossing.)

family member: ______________________

signature: ______________________

for the child

meet the wind turbine



your wind turbine – it spins in the wind and lights up an led.

On a windy hill, Meera watches a giant white windmill turn, slow and steady. Her teacher says it isn't grinding grain – it's making electricity for the whole village. Meera wonders: how can spinning blades light up a bulb? today, you'll find out.

think about it

where have you seen something spin in the wind? what do you think that spinning could power?

words to know

energy

led

generator

did you know?

a spinning motor can **make** electricity – then it's called a **generator**.

for the child

🔧 building & making

build your wind turbine.

this is what you're building.

📖 **use your model manual**

follow your **wind turbine manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: connect the motor to the led block, so it **spins in the wind and lights up an led**.

wind turbine

for the child

🔧 building & making

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.

label these four parts

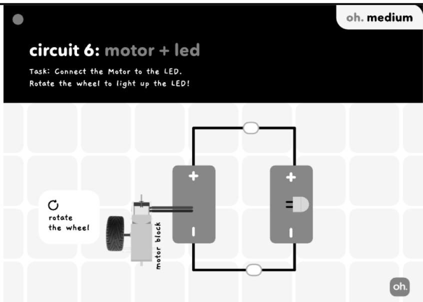
the blades | the motor | the led | the wires

⭐ **bonus:** draw an arrow to show how it moves.

wind turbine

observing & understanding

build this circuit – then test it like a scientist.



your test

spin the blades at three speeds. rank the led brightness from 1 (dimmest) to 3 (brightest).

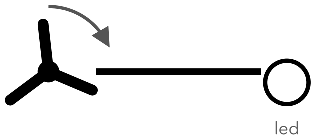
speed	brightness rank (1-3)

✎ what does your table show? write one sentence.

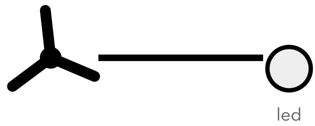
✎

🔬 observing & understanding

the big idea – why does it work?



spinning fast



barely spinning

the spinning motor is a **generator**: motion in → electricity out. a faster spin makes a higher **voltage**, so more current flows and the led is brighter. an led has **polarity** – it only lets current through one way, so it must face the right way to light.

✎ in your own words

explain the big idea to a friend – write it in one or two sentences.

✎

for the child

problem solving

design challenge

real engineers try more than one idea, then compare.

1 · design a turbine that lights the led in the LOWEST wind

list two changes you could make.

2 · predict which change helps more, then test it

write your prediction, then the result.

3 · keep only the change that worked

draw your best design.

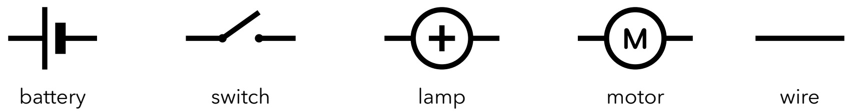
✎ draw and label your designs here

problem solving

engineer challenge – draw the circuit diagram

draw a **circuit diagram**: the generator (motor) connected to the led, showing which way the led must face.

use these circuit symbols



✎ draw your circuit as a proper diagram – a **complete loop** using the symbols above.

✎ draw your circuit diagram here

for the child

🗣️ presenting

be the engineer – present your machine

show your wind turbine to the group and take them through it:

1 **present** – present your wind turbine to the group – show it working and name what each block does.

2 **walk through** – walk them through how it works – trace how the electricity flows, step by step, so they can follow.

3 **justify** – explain **why** it's built this way – what each block adds, and what would happen if you removed or swapped one.

4 **field it** ⭐ – take a question from the audience – answer a 'what if…?' – or coach a friend to build or fix the same circuit.

🏠 at-home show & tell

take it home and show a family member. explain: **what problem does it solve? how does it work?** (it turns wind into electricity to light a lamp.)

family member: ______________________________

signature: ______________________________

for the child

meet the soccer bot

your soccer bot – it drives forward and back when you flip the switch.

It's the big match! Kabir grips his controller as his little robot races across the floor, nudges the ball, and – GOAL! But to defend, he has to make it reverse, fast. how can one robot go both forwards and backwards? that's your challenge.

think about it

think of a toy that moves forward and back. how do you make it change direction?

words to know

motor · direction · reverse

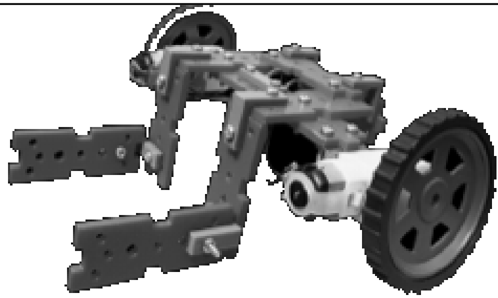
did you know?

swap the two wires on a motor and it spins the **other** way – that's **reversing**.

👩 for the child

🔧 building & making

build your soccer bot.



this is what you're building.

📖 **use your model manual**

follow your **soccer bot manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: wire the motors through the dpdt switch, so it **drives forward and back when you flip the switch**.

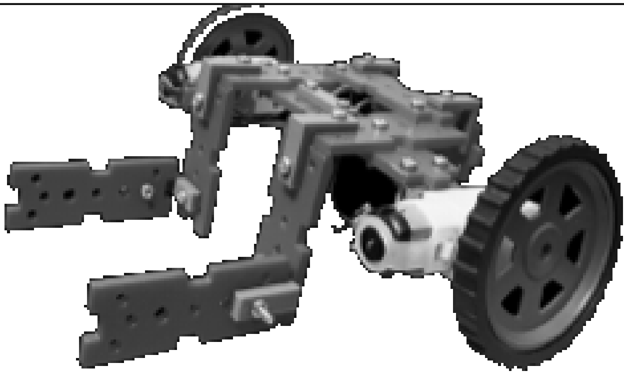
soccer bot

🔧 building & making

for the child

⭐ name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

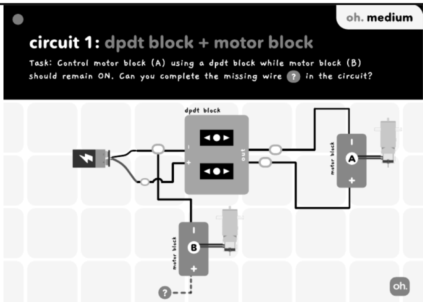
the motor | the wheels | the battery | the switch

⭐ **bonus:** draw an arrow to show how it moves.

for the child

observing & understanding

build this circuit – then test it like a scientist.



your test

flip the dpdt switch four times. record the switch position and which way the wheels turned.

flip	switch position	wheel direction
1		
2		
3		
4		

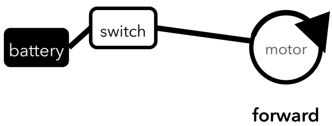
✎ what does your table show? write one sentence.

✎

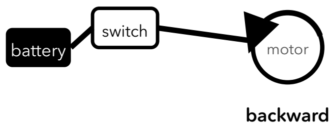
for the child

🔬 observing & understanding

the big idea – why does it work?



switch one way



switch the other way

a motor's direction depends on which way the **current** flows through it. swapping the two connections reverses the current, so the motor reverses. a **dpdt switch** swaps both connections at once for you.

✎ in your own words

explain the big idea to a friend – write it in one or two sentences.

✎

soccer bot

for the child

problem solving

design challenge

real engineers try more than one idea, then compare.

1 · design controls so the bot turns LEFT and RIGHT

how many motors and switches do you need? draw it.

2 · work out what each switch position should do

fill a small table of your own.

3 · pick the simplest design that still turns both ways

circle it.

✎ draw and label your designs here

for the child

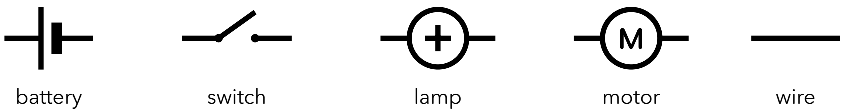
soccer bot

problem solving

engineer challenge – draw the circuit diagram

draw a **circuit diagram**: battery → dpdt switch → motor, and note what each switch position does.

use these circuit symbols



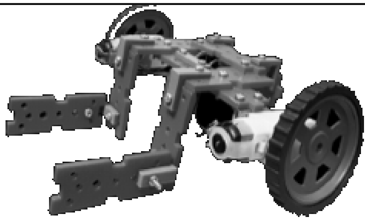
✎ draw your circuit as a proper diagram – a **complete loop** using the symbols above.

✎ draw your circuit diagram here

for the child

🗣️ presenting

be the engineer – present your machine



show your soccer bot to the group and take them through it:

1 **present** – present your soccer bot to the group – show it working and name what each block does.

2 **walk through** – walk them through how it works – trace how the electricity flows, step by step, so they can follow.

3 **justify** – explain **why** it's built this way – what each block adds, and what would happen if you removed or swapped one.

4 **field it** ⭐ – take a question from the audience – answer a 'what if…?' – or coach a friend to build or fix the same circuit.

🏠 at-home show & tell

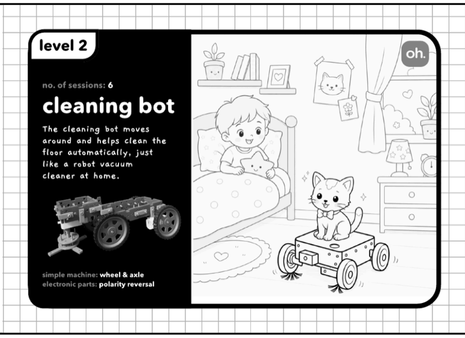
take it home and show a family member. explain: **what problem does it solve? how does it work?** (you drive it to chase the ball and score goals.)

family member: ______________________________

signature: ______________________________

for the child

meet the cleaning bot



your cleaning bot – it drives along and sweeps at the same time.

Diya's room is a mess again! She wishes for a little robot that could drive around AND sweep at the same time. Her uncle smiles: one battery can do more than one job at once. how? today, you'll build it and see.

think about it

if a robot could do two jobs at the same time, which two jobs would you give it?

words to know

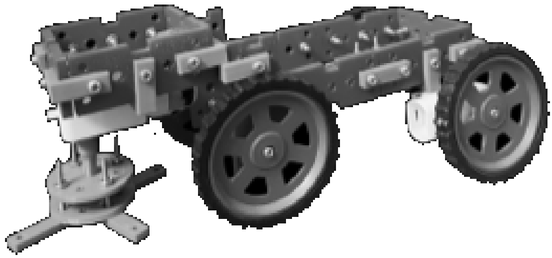
power · parallel · current

did you know?

when one battery runs two things at once, we say they are wired in **parallel**.

🔧 building & making

build your cleaning bot.



this is what you're building.

📖 use your model manual

follow your **cleaning bot manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: wire both motors to the one battery, so it **drives along and sweeps at the same time**.

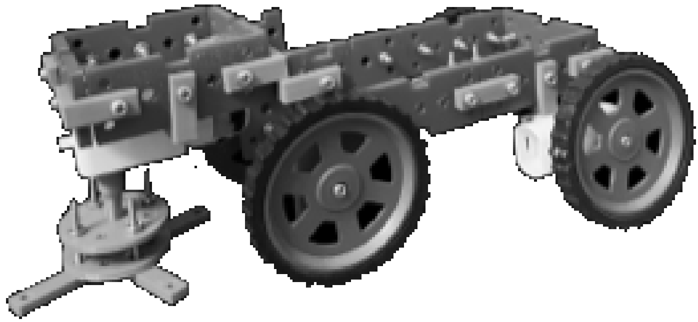
for the child

cleaning bot

building & making

name the parts

on the picture of your model, circle each part and write its name next to it.



label these four parts

the wheels' motor | the brush's motor | the battery | the brush

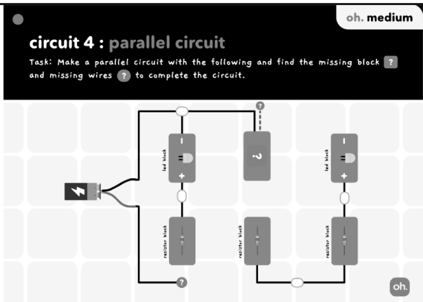
bonus: draw an arrow to show how it moves.

cleaning bot

for the child

observing & understanding

build this circuit – then test it like a scientist.



your test

run only the wheels, then only the brush, then both. does the brush slow down when both run? note what you see.

what's running	brush speed (fast/ok/slow)

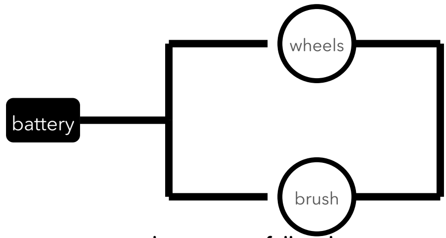
✎ what does your table show? write one sentence.

✎

for the child

🔬 observing & understanding

the big idea – why does it work?



one battery, two full paths

wiring the two motors in **parallel** gives each its own full path to the battery, so both get full power. wired in **series** (one after the other) they would share the power and both run weaker.

✎ in your own words

explain the big idea to a friend – write it in one or two sentences.

✎

cleaning bot

for the child

problem solving

design challenge

real engineers try more than one idea, then compare.

1 · design a bot that runs wheels + brush + a light, all from one battery
draw it wired in parallel.

2 · work out how many paths leave the battery
label them on your drawing.

3 · compare parallel vs series for your bot
which keeps everything strong? say why.

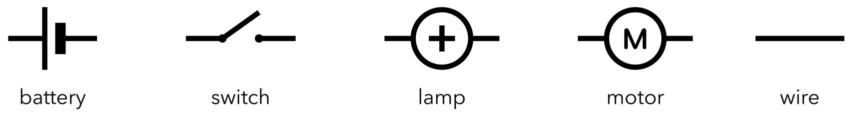
✎ draw and label your designs here

problem solving

engineer challenge – draw the circuit diagram

draw a **circuit diagram**: one battery with two parallel branches – a wheels motor and a brush motor.

use these circuit symbols

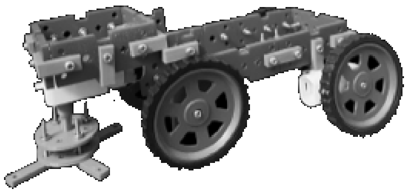


✎ draw your circuit as a proper diagram – a **complete loop** using the symbols above.

✎ draw your circuit diagram here

🗣️ presenting

be the engineer – present your machine



show your cleaning bot to the group and take them through it:

1. **present** – present your cleaning bot to the group – show it working and name what each block does.
2. **walk through** – walk them through how it works – trace how the electricity flows, step by step, so they can follow.
3. **justify** – explain **why** it's built this way – what each block adds, and what would happen if you removed or swapped one.
4. **field it** ⭐ – take a question from the audience – answer a 'what if…?' – or coach a friend to build or fix the same circuit.

🏠 at-home show & tell

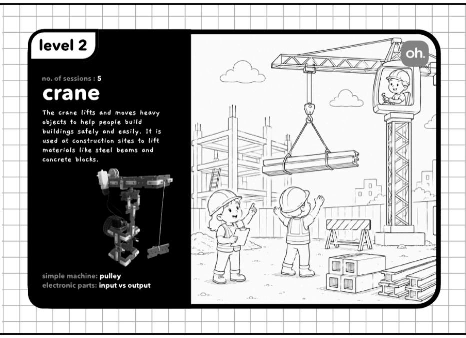
take it home and show a family member. explain: **what problem does it solve? how does it work?** (it sweeps the floor while it drives around.)

family member: ______________________________

signature: ______________________________

for the child

meet the sensor-controlled crane



your sensor-controlled crane — it lifts the load when the sensor is triggered.

At the port, huge cranes lift heavy boxes off the ships. Rehan notices something clever — the crane seems to KNOW when a box is under it, and only then does it lift. how can a machine sense and then act all by itself? let's build one.

💭 think about it

where have you seen a machine that notices something and then acts by itself — like a door that opens as you walk up?

words to know

sensor

input

output

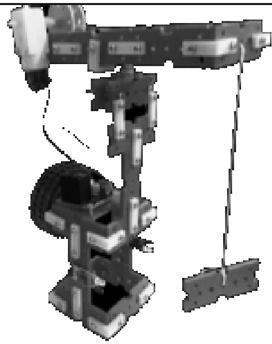
💡 did you know?

a machine part that **senses** the world (like light or an object) is called a **sensor**.

for the child

🔧 building & making

build your sensor-controlled crane.



this is what you're building.

📖 **use your model manual**

follow your **sensor-controlled crane manual** to build it from your kit – one stage a day. build it, test it, then improve it as you go.

☐ **tick when you've built it**

✎ next: wire the sensor to the motor, so it **lifts the load when the sensor is triggered**.

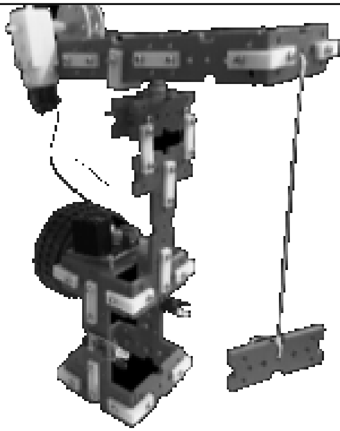
for the child

sensor-controlled crane

building & making

name the parts

✎ on the picture of your model, circle each part and write its name next to it.



label these four parts

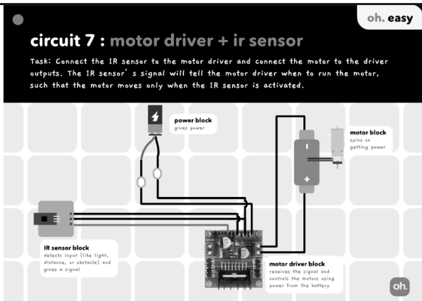
the sensor | the motor | the hook | the battery

bonus: draw an arrow to show how it moves.

for the child

observing & understanding

build this circuit – then test it like a scientist.



your test

move an object toward the sensor. measure the distance where the motor starts. try 3 times.

try	distance it triggers (cm)
1	
2	
3	

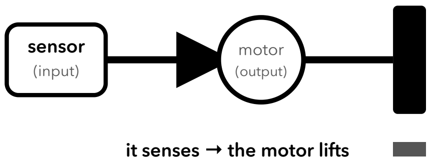
✎ what does your table show? write one sentence.

✎

for the child

observing & understanding

the big idea – why does it work?



the sensor is an **input** – it detects something and sends a **signal**. the motor is the **output** – it acts on that signal. every smart machine works this way: **input → decide → output**.

✎ in your own words

explain the big idea to a friend – write it in one or two sentences.

✎

sensor-controlled crane

for the child

problem solving

design challenge

real engineers try more than one idea, then compare.

1 · design a crane that lifts when a box is near AND stops at the top
what inputs and outputs do you need?

2 · draw the input → output flow for your crane
label each part.

3 · add one more sensor to make it safer
say what it would sense.

✎ draw and label your designs here

for the child

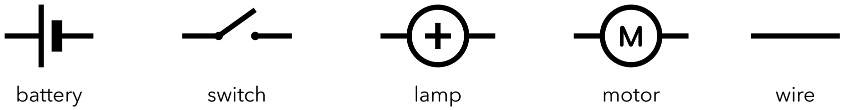
sensor-controlled crane

problem solving

⭐ engineer challenge – draw the circuit diagram

draw a **block diagram**: sensor (input) → controller → motor (output).

use these circuit symbols



✎ draw your circuit as a proper diagram – a **complete loop** using the symbols above.

✎ draw your circuit diagram here

for the child

presenting

be the engineer – present your machine

show your sensor-controlled crane to the group and take them through it:

1 **present** – present your sensor-controlled crane to the group – show it working and name what each block does.

2 **walk through** – walk them through how it works – trace how the electricity flows, step by step, so they can follow.

3 **justify** – explain **why** it's built this way – what each block adds, and what would happen if you removed or swapped one.

4 **field it** ⭐ – take a question from the audience – answer a 'what if…?' – or coach a friend to build or fix the same circuit.

at-home show & tell

take it home and show a family member. explain: **what problem does it solve? how does it work?** (it senses a load and lifts it, all by itself.)

family member: ____________________

signature: ____________________